JLPT AI App, Team Roles

Each member works independently with fake or sample data, then integrates step by step

HP: Backend and Infrastructure

Builds the foundation of the project and the server every screen talks to.

Independent work (starts week 1)

- GitHub organization, repository, branch protection and Project board
- Docker and docker-compose with the database
- Database tables (users, sessions, units, words, grammar, questions, results)
- Google login verification and user accounts
- API endpoints exactly as written in the API contract, first returning fake data, then real data
- Study plan calculation (units left divided by days left)
- Load ANK's sample unit into the database so the API returns real content early

Doesn't wait for anyone because

- The API contract defines what to build, and the sample unit gives real test data

Integration points

- LMHP switches the app from fake data to the real API
- SWYH's AI code plugs into backend/ai/
- ANK's content files get loaded into the database

Later sprint

- Push notifications, automatic checks (GitHub Actions), cloud deployment

LMHP: Frontend and Design

Builds everything the user sees and touches, on iOS, Android and web (Expo).

Independent work (starts week 1)

- Design system: colors, fonts, spacing, icons, logo
- Screen sketches for all main screens
- Build every screen using a fake data file shaped exactly like the API contract:
- Login screen
- First-time setup form with input checks (score 0 to 180, whole numbers only, official exam dates)
- Dashboard with session cards, progress bar and pass prediction
- Session and unit screens with meanings and example sentences
- Quiz screen with green and red feedback
- Mock test screen with timer and review screen
- Settings screen
- Test Burmese and Japanese text display on real phones

Doesn't wait for anyone because

- The fake data file has the same shape as the real API, so screens work before the backend exists

Integration points

- Replace fake data with real API calls once HP's endpoints are ready (one screen at a time)
- Connect the Google login button to HP's login check
- Place SWYH's chat screen into the app

ANK: Data and Content

Prepares the learning material and makes sure everything users study is correct.

Independent work (starts week 1)

- Write one complete sample unit by hand (20 words, 1 grammar pattern, meanings, example sentences, Burmese translations, 5 quiz questions). This becomes the template for SWYH's scripts and test data for HP and LMHP.
- Choose the source word list and grammar list, and note it in docs/
- Plan which words and grammar go into which session and unit, ordered from easier to harder
- Prepare the proverb (kotowaza) list with Burmese meanings
- Write the mock test template per level from the official JLPT format (question types, counts, time limits)

Doesn't wait for anyone because

- Planning, the sample unit, proverbs and mock templates need no code from others

Integration points

- Review AI-drafted content as SWYH's scripts produce it, mark items as checked or flag them
- Test screens as LMHP finishes them and report problems as GitHub Issues
- After December 6, take over the database tables from HP

Exam protection (until December 6)

- Only review and planning work, no large writing tasks

SWYH: AI Integration

Builds everything in the app that uses AI, from generating content to answering users' questions.

Independent work (starts week 1)

- Compare LLM options on Burmese quality, Japanese accuracy and cost, and pick one with the team
- Set up the API key in .env and basic error handling
- Using ANK's sample unit as the target format, build a script that drafts example sentences, readings and Burmese meanings
- Build a script that drafts quiz questions with 4 options
- Build a script that generates explanations for every answer option
- Build the "Ask AI" chatbot as a standalone script first (question in, answer out), using the sample unit as its RAG data
- Build the chat screen following LMHP's design system
- Create a test set of tricky questions and measure explanation accuracy
- Keep a prompt log of what improved results

Doesn't wait for anyone because

- The sample unit gives real data and the content format defines the output, so the scripts run on their own

Integration points

- Send drafted content to ANK for review
- Move chatbot code into backend/ai/ and connect it through an endpoint agreed with HP
- Store generated explanations in HP's database
- Give LMHP the chat screen to place in the app
- Add study advice once HP's study plan endpoint is ready

Team Schedule

Two agreements come first, so nobody has to wait for anybody:

- **API contract** (HP leads, LMHP and SWYH join): which requests exist and what JSON each one returns
- **Content format** (ANK and SWYH): the exact fields for a word, grammar pattern, example sentence and quiz question, plus one sample unit

Timeline

- **Week 1**: agreements, GitHub and Docker setup, sample unit
- **Weeks 2 to 4**: everyone builds independently with fake or sample data
- **Week 5**: first integration, login plus dashboard plus one unit working end to end
- **Week 6 onward**: connect the rest feature by feature, each through a pull request

Rules for everyone

- If the API contract or content format needs to change, tell the team first, because everyone builds against it
- Work on your own branch, never push directly to main
- Every change goes through a pull request and one teammate's review
- Stay inside your own folder, and ask before changing someone else's area
- Never commit API keys or the .env file
- Use AI to help you code, but you must be able to explain everything you commit